Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
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Project Title:	Effects of Microgravity on Microbial Physiology		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98021	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	93-OLMSA-07
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	Shuttle		
Flight Assignment:	STS-107		
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	None		
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Performance Goal Text:			

Microorganisms are capable of quickly adapting to changes in their environment. Metabolic pathways adjust to new environmental conditions to optimize growth and ensure survival. Environmental adaptation is a key tool used in microbial research to investigate the basic physiology of microbial species. Temperature, pH, and growth media constituent manipulation are classic examples of environmental parameter control schemes used to induce observable metabolic changes. While classic induced physiology adaptation has been studied extensively, the effects of gravity on microbial physiology remain essentially unknown. Prior space flight research indicates that exposure to microgravity can lead to increased growth rates and increased resistance to antimicrobial agents. However, the limited data and conflicting results of this body of work demand that a well-controlled, prolific methodology be adapted for microbial space flight research.
The hypothesis to be tested here is that microgravity will induce functional changes in microbial physiology. To identify

Task Description:	and evaluate these changes, a two-phase project will be conducted to first screen for and then target statistically significant differences in physiological behavior between ground-based and on-orbit microbial cultures. The screening phase of the project will investigate the growth, substrate utilization characteristics, and reaction to antimicrobial agents of twelve microbial strains including bacteria and yeasts. These microbes will be analyzed simultaneously on-orbit and on the ground using an automated microbial analysis technology, the Vitek System manufactured by bioMerieux Vitek, Inc. Microbial growth patterns will be measured in standard Vitek identification and antibiotic susceptibility test cards. A physiological profile consisting of observed growth rates, substrate utilization characteristics, and antimicrobial susceptibilities will be obtained for each microbial strain. Physiological profiles from the terrestrial and on-orbit microbial analyses will be compared to determine statistically significant differences in behavior. In the second phases of the project, Vitek test cards containing unique media formulations and/or antimicrobial agents will be developed to further investigate each of the specific metabolic differences identified in the screening phase. These test cards will be analyzed in subsequent shuttle flights to elucidate the cellular mechanisms responsible for the observed metabolic anomalies. This study will provide a basic understanding of the effects of microgravity on microbial physiology. In addition, the antimicrobial physiological profile data will aid in the development of effective therapeutic treatment for controlling on-orbit infectious disease. Adaptation of the Vitek instrument for space flight will provide the first ever on-orbit comprehensive microbial identification capability, an invaluable capability for clinical and environmental microbiology on future long-duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Microbial Physiology Flight Experiment (MPFE) is aimed at identifying gravity-dependent physiological processes by comparing the growth and metabolism of a broad range of microorganisms cultured in space and on the ground. The anticipated results will indicate which microbes and specifically which metabolic pathways are affected by variation in the gravitational environment. These results will benefit the fields of biology, clinical microbiology, and applied microbiology. From a basic biological research perspective, identification of gravity- dependent physiological processes will add significantly to our overall understanding of how the environment, and gravity in particular, affects the basic processes of life. Bacteria and yeast are some of the simplest forms of life, yet they exhibit physiological pathways that have been conserved through millions of years of evolution. In this way, microbes represent the ideal model for studying the most basic life processes. The simplicity of these organisms allows for targeting of specific bioprocesses without the complications of intercellular and system-level interactions. Gravity represents an under- utilized tool in the array of environmental parameters used to elicit metabolic responses. Gravity is unique in this array in that it is the one environmental constant under which all forms of life evolved. For this reason, it will be especially intriguing to observe how microbes adapt to an environmental parameter which their genetic programming has never encountered. One might suggest that observed microbial adaptive behaviors may be extrapolated to help explain observed adaptive physiological phenomena in humans as well. Clinical microbiology is a field which may benefit significantly from the MPFE project. Prior work has indicated that microbes exhibit an increase in resistance to antimicrobial agents when grown in microgravity. If this behavior can be confirmed, the flight experiment will provide an excellent opportunity to investigate in detail the mechanisms of antibiotic resistance. The growing prevalence of resistant pathogen strains has made antimicrobial research of critical importance to the health of the public as a whole. Results of the MPFE project will provide a unique set of data for the clinical microbiology community to use in combating infectious disease. Gravity-dependent bioprocesses are of considerable interest in the field of applied microbiology as well. Biochemists and crystallographers currently take advantage of the minimal gravity on the Shuttle in Earth orbit to produce crystals that would otherwise not be possible. Physicists researching combustion and phase-change phenomena utilize the unique conditions of microgravity to investigate processes that cannot occur here on Earth. What about biological processes? Biological gravitational effects are not so easily anticipated as those for physical phenomena, yet it is these effects which one hopes will make space processing a valuable commodity. This project will constitute the first steps in the process of identifying those microgravity-induced bioprocesses which may be harnessed to produce unique biochemicals and pharmaceuticals in space.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)